Digital Logic and Design

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Lecture No. 09

Recap

- Karnaugh Maps
- Mapping Standard SOP expressions
- Mapping Non-Standard SOP expressions
- Simplification of K-maps
- Don’t care states

Mapping a Standard POS expression

- Selecting n-variable K-map
- 0 marked in cell for each maxterm
- Remaining cells marked with 1

Mapping of Standard POS expression

- POS expression

$$(A+B+\overline{C}).(A+\overline{B}+C).(\overline{A}+B+\overline{C}).(\overline{A}+\overline{B}+\overline{C})$$

Mapping of Standard POS expression

- POS expression

$$(A + B + \overline{C}).(A + \overline{B} + C).(\overline{A} + B + \overline{C}).(\overline{A} + \overline{B} + \overline{C})$$

AB\C	0	1
00	1	0
01	0	1
11	1	0
10	1	0

A\BC	00	01	11	10
0	1	0	1	0
1	1	0	0	1

Simplification of POS expressions using K-map

- Mapping of expression
- Forming of Groups of 0s
- Each group represents sum term
- 3-variable K-map
  - 1 cell group yields a 3 variable sum term
  - 2 cell group yields a 2 variable sum term
  - 4 cell group yields a 1 variable sum term
  - 8 cell group yields a value of 0 for function

Simplification of POS expressions using K-map3

- 4-variable K-map
  - 1 cell group yields a 4 variable sum term
  - 2 cell group yields a 3 variable sum term
  - 4 cell group yields a 2 variable sum term
  - 8 cell group yields a 1 variable sum term
  - 16 cell group yields a value of 0 for function

Simplification of POS expressions using K-map

$$(B+C).(A+\overline{B}+\overline{C})$$

AB\C	0	1
00	0	1
01	1	0
11	1	1
10	0	1

A\BC	00	01	11	10
0	0	1	1	1
1	1	0	0	0

$$(A+B+C).(\overline{A}+\overline{C}).(\overline{A}+\overline{B})$$

Simplification of POS expressions using K-map

$$(B+C).(A+\overline{B}+\overline{C})$$

AB\C	0	1
00	0	1
01	1	0
11	1	1
10	0	1

A\BC	00	01	11	10
0	0	1	1	1
1	1	0	0	0

$$(A+B+C).(\overline{A}+\overline{C}).(\overline{A}+\overline{B})$$

Simplification of POS expressions using K-map

$$(A+B).(B+C)$$

AB\C	0	1
00	0	0
01	1	1
11	1	1
10	0	1

A\BC	00	01	11	10
0	0	0	1	1
1	1	1	1	0

$$(A+B).(\overline{A}+\overline{B}+C)$$

Simplification of POS expressions using K-map

$$(A+\overline{B}+C).(A+C+D).(B+\overline{C}+D)$$

AB\CD	00	01	11	10
00	0	1	1	0
01	0	0	1	1
11	1	1	1	1
10	1	1	1	0

Simplification of POS expressions using K-map

$$(A + C).(C + \overline{D}).(B + \overline{C} + D)$$

AB\CD	00	01	11	10
00	0	0	1	0
01	0	0	1	1
11	1	0	1	1
10	1	0	1	0

Simplification of POS expressions using K-map

$$(A+\overline{B}+C).(A+\overline{B}+\overline{D}).(B+C+\overline{D}).(\overline{A}+\overline{B}+\overline{C}+D)$$

AB\CD	00	01	11	10
00	1	0	1	1
01	0	0	0	1
11	1	1	1	0
10	1	0	1	1

Conversion between SOP & POS using K-map

- Groups of 1s represents SOP expression
- Groups of 0s represents POS expression

Conversion between SOP & POS using K-map

$$\overline{B}\,\overline{D} + \overline{B}C + AB\overline{C} + ABD + \overline{A}C\overline{D}$$

AB\CD	00	01	11	10
00	1	0	1	1
01	0	0	0	1
11	1	1	1	0
10	1	0	1	1

5-Variable K-map

- Represented as two, 4 variable K-map

5-Varaible K-map

A $= 0$

BC\DE	00	01	11	10
00	0	1	3	2
01	4	5	7	6
11	12	13	15	14
10	8	9	11	10

5-Varaible K-map

A $= 1$

BC\DE	00	01	11	10
00	16	17	19	18
01	20	21	23	22
11	28	29	31	30
10	24	25	27	26

Simplification of a 5-Variable K-map

- 5 variable K-map mapping
- 5 variable K-map grouping
- 5 variable K-map simplification

5-Varaible K-map simplification

A $= 0$

BC\DE	00	01	11	10
00	0	1	0	1
01	0	1	0	0
11	0	0	0	1
10	0	0	1	1

5-Varaible K-map simplification

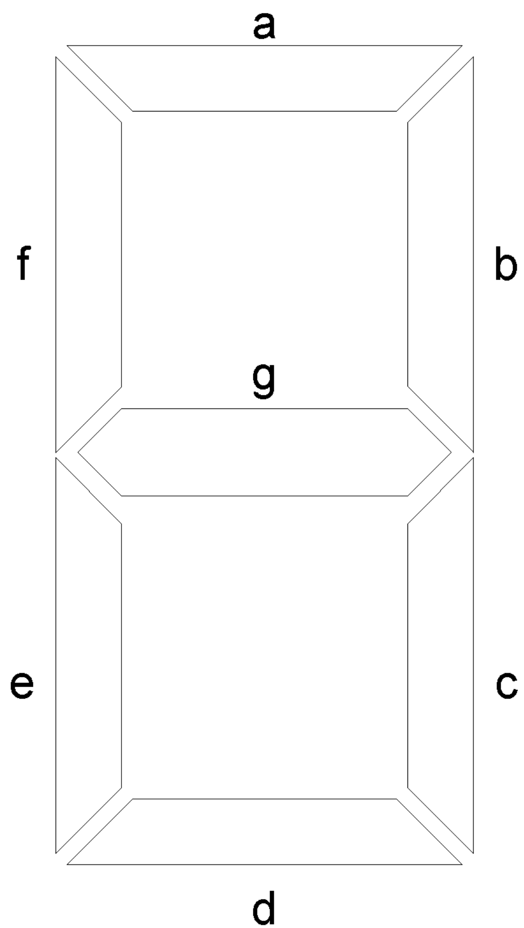
A $= 1$

BC\DE	00	01	11	10
00	1	1	0	0
01	1	1	0	0
11	0	0	0	1
10	0	1	1	1

Functions having multiple outputs

- Ckt receives a BCD number input
- Displays decimal number 0 to 9 on a single digit 7-segment display
- Ckt receives two 2-bit numbers A and B
- Sets one of three outputs to >, =, or <

7-Segment Display



Function Table for Segment ‘a’

Inputs				Output
A	B	C	D	a
0	0	0	0	1
0	0	0	1	0
0	0	1	0	1
0	0	1	1	1
0	1	0	0	0
0	1	0	1	1
0	1	1	0	1
0	1	1	1	1

Inputs				Output
A	B	C	D	a
1	0	0	0	1
1	0	0	1	1
1	0	1	0	X
1	0	1	1	X
1	1	0	0	X
1	1	0	1	X
1	1	1	0	X
1	1	1	1	X

Function Table for Segment ‘b’

Inputs				Output
A	B	C	D	b
0	0	0	0	1
0	0	0	1	1
0	0	1	0	1
0	0	1	1	1
0	1	0	0	1
0	1	0	1	0
0	1	1	0	0
0	1	1	1	1

Inputs				Output
A	B	C	D	b
1	0	0	0	1
1	0	0	1	1
1	0	1	0	X
1	0	1	1	X
1	1	0	0	X
1	1	0	1	X
1	1	1	0	X
1	1	1	1	X

Function Table for Segment ‘c’

Inputs				Output
A	B	C	D	c
0	0	0	0	1
0	0	0	1	1
0	0	1	0	0
0	0	1	1	1
0	1	0	0	1
0	1	0	1	1
0	1	1	0	1
0	1	1	1	1

Inputs				Output
A	B	C	D	c
1	0	0	0	1
1	0	0	1	1
1	0	1	0	X
1	0	1	1	X
1	1	0	0	X
1	1	0	1	X
1	1	1	0	X
1	1	1	1	X

Function Table for Segment ‘d’

Inputs				Output
A	B	C	D	d
0	0	0	0	1
0	0	0	1	0
0	0	1	0	1
0	0	1	1	1
0	1	0	0	0
0	1	0	1	1
0	1	1	0	1
0	1	1	1	0

Inputs				Output
A	B	C	D	d
1	0	0	0	1
1	0	0	1	1
1	0	1	0	X
1	0	1	1	X
1	1	0	0	X
1	1	0	1	X
1	1	1	0	X
1	1	1	1	X

Function Table for Segment ‘e’

Inputs				Output
A	B	C	D	e
0	0	0	0	1
0	0	0	1	0
0	0	1	0	1
0	0	1	1	0
0	1	0	0	0
0	1	0	1	0
0	1	1	0	1
0	1	1	1	0

Inputs				Output
A	B	C	D	e
1	0	0	0	1
1	0	0	1	0
1	0	1	0	X
1	0	1	1	X
1	1	0	0	X
1	1	0	1	X
1	1	1	0	X
1	1	1	1	X

Function Table for Segment 'f'

Inputs				Output
A	B	C	D	f
0	0	0	0	1
0	0	0	1	0
0	0	1	0	0
0	0	1	1	0
0	1	0	0	1
0	1	0	1	1
0	1	1	0	1
0	1	1	1	0

Inputs				Output
A	B	C	D	f
1	0	0	0	1
1	0	0	1	1
1	0	1	0	X
1	0	1	1	X
1	1	0	0	X
1	1	0	1	X
1	1	1	0	X
1	1	1	1	X

Function Table for Segment ‘g’

Inputs				Output
A	B	C	D	g
0	0	0	0	0
0	0	0	1	0
0	0	1	0	1
0	0	1	1	1
0	1	0	0	1
0	1	0	1	1
0	1	1	0	1
0	1	1	1	0

Inputs				Output
A	B	C	D	g
1	0	0	0	1
1	0	0	1	1
1	0	1	0	X
1	0	1	1	X
1	1	0	0	X
1	1	0	1	X
1	1	1	0	X
1	1	1	1	X

Karnaugh Map for Segment ‘a’

$$A + C + BD + \overline{B}\,\overline{D}$$

AB\CD	00	01	11	10
00	1	0	1	1
01	0	1	1	1
11	x	x	x	x
10	1	1	x	x

Karnaugh Map for Segment ‘b’

$$\overline{B} + \overline{C}\overline{D} + CD$$

AB\CD	00	01	11	10
00	1	1	1	1
01	1	0	1	0
11	x	x	x	x
10	1	1	x	x

Karnaugh Map for Segment ‘c’

$$\overline{C} + D + B$$

AB\CD	00	01	11	10
00	1	1	1	0
01	1	1	1	1
11	x	x	x	x
10	1	1	x	x

Karnaugh Map for Segment 'd'

$$A + \overline{B}\,\overline{D} + \overline{B}C + C\overline{D} + B\overline{C}D$$

AB\CD	00	01	11	10
00	1	0	1	1
01	0	1	0	1
11	x	x	x	x
10	1	1	x	x

Karnaugh Map for Segment 'e'

$$\overline{B}\,\overline{D} + C\overline{D}$$

AB\CD	00	01	11	10
00	1	0	0	1
01	0	0	0	1
11	x	x	x	x
10	1	0	x	x

Karnaugh Map for Segment ‘f’

$$B + \overline{C}\,\overline{D} + B\overline{C} + B\overline{D}$$

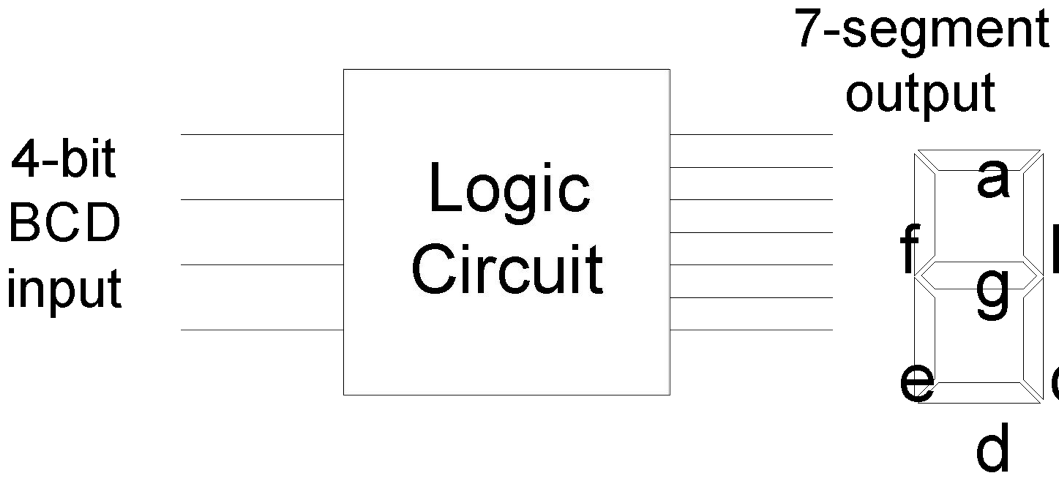
AB\CD	00	01	11	10
00	1	0	0	0
01	1	1	0	1
11	x	x	x	x
10	1	1	x	x

Karnaugh Map for Segment ‘g’

$$A + B\overline{C} + C\overline{D} + \overline{B}C$$

AB\CD	00	01	11	10
00	0	0	1	1
01	1	1	0	1
11	x	x	x	x
10	1	1	x	x

7-Segment Circuit



Comparator Circuit

- Inputs two 2-bit binary numbers A and B
- Has three outputs
- A>B
- A=B
- A<B

Function Table for A>B

Inputs				Output
A_1	A_0	B_1	B_0	A>B
0	0	0	0	0
0	0	0	1	0
0	0	1	0	0
0	0	1	1	0
0	1	0	0	1
0	1	0	1	0
0	1	1	0	0
0	1	1	1	0

Inputs				Output
A_1	A_0	B_1	B_0	A>B
1	0	0	0	1
1	0	0	1	1
1	0	1	0	0
1	0	1	1	0
1	1	0	0	1
1	1	0	1	1
1	1	1	0	1
1	1	1	1	0

Function Table for A=B

Inputs				Output
A_1	A_0	B_1	B_0	A=B
0	0	0	0	1
0	0	0	1	0
0	0	1	0	0
0	0	1	1	0
0	1	0	0	0
0	1	0	1	1
0	1	1	0	0
0	1	1	1	0

Inputs				Output
A_1	A_0	B_1	B_0	A=B
1	0	0	0	0
1	0	0	1	0
1	0	1	0	1
1	0	1	1	0
1	1	0	0	0
1	1	0	1	0
1	1	1	0	0
1	1	1	1	1

Function Table for A<B

Inputs				Output
A_1	A_0	B_1	B_0	A<B
0	0	0	0	0
0	0	0	1	1
0	0	1	0	1
0	0	1	1	1
0	1	0	0	0
0	1	0	1	0
0	1	1	0	1
0	1	1	1	1

Inputs				Output
A_1	A_0	B_1	B_0	A<B
1	0	0	0	0
1	0	0	1	0
1	0	1	0	0
1	0	1	1	1
1	1	0	0	0
1	1	0	1	0
1	1	1	0	0
1	1	1	1	0

Karnaugh Map for A>B

$$A_1\overline{B_1} + A_0\overline{B_1}\,\overline{B_0} + A_1A_0\overline{B_0}$$

A_1A_0/ B_1B_0	00	01	11	10
00	0	0	0	0
01	1	0	0	0
11	1	1	0	1
10	1	1	0	0

Karnaugh Map for A=B

$$\overline{A_1}\,\overline{A_0}\,\overline{B_1}\,\overline{B_0} + \overline{A_1}A_0\overline{B_1}B_0 + A_1A_0B_1B_0 + A_1\overline{A_0}B_1\overline{B_0}$$

A_1A_0/ B_1B_0	00	01	11	10
00	1	0	0	0
01	0	1	0	0
11	0	0	1	0
10	0	0	0	1

Karnaugh Map for A<B

$$\overline{A_1}B_1 + \overline{A_1}\,\overline{A_0}B_0 + \overline{A_0}B_1B_0$$

A_1A_0/ B_1B_0	00	01	11	10
00	0	1	1	1
01	0	0	1	1
11	0	0	0	0
10	0	0	1	0

Odd Prime Number Circuit

- Circuit detects odd prime numbers for a 5-bit input number
- Function represented by minterms

  3, 5, 7, 11, 13, 17, 19, 23, 29 and 31